

- DP-600 Fabric Analytics Engineer
- DP-203 Azure Data Engineer
- PL-300 Power BI Analyst
- AI-900 Azure AI Fundamentals
- DP-900 Data Fundamentals
- AZ-900 Azure Fundamentals

PROFESSIONAL SUMMARY

Strategic Lead Data Engineer with **8+ years of experience** and a proven **5-year track record** of leading high-performance engineering teams and managing direct stakeholder engagements. Expert in designing scalable **Lakehouse architectures on Microsoft Azure and Microsoft Fabric**, acting as the primary techno-functional bridge between business goals and technical delivery. Specializes in **Cloud FinOps**, significantly reducing operational costs through advanced compute optimization (DIUs, Parallelism) and SLA optimization. Deep domain expertise in the **EU BFSI sector**, ensuring strict GDPR / data sovereignty compliance while delivering rapid innovation and **99.9% system reliability**.

CORE TECHNICAL COMPETENCIES

- Core Architecture:** Medallion Architecture, Data Mesh, Cloud FinOps, Disaster Recovery (DR), Governance, Stakeholder Management, Kimball Modeling, Inmon Methodology, Data Vault 2.0, Star / Snowflake Schemas, SCDs, Zero-Copy Architecture
- Big Data & Azure:** Microsoft Fabric, OneLake & Direct Lake, Azure Synapse Analytics, Azure Data Factory, ADLS Gen2, Stream Analytics, Event Hubs, Databricks, Azure SQL, Azure Cosmos DB, Logic Apps, Azure Monitor / KQL
- ETL / ELT & Pipelines:** ADF Mapping Data Flows, Idempotent Pipeline Design, Apache Airflow, SCD Type 2, ACID Compliance, Delta Lake, Liquid Clustering, Z-Ordering, Copy-On-Write (CoW), MERGE INTO / UPSERT, Delta Time Travel, Data Orchestration
- DevOps & Governance:** Azure DevOps, CI/CD (YAML), Row-Level Security (RLS), GDPR Compliance, Data Sovereignty, Azure Key Vault, Service Principals, OAuth Handshake, Entra ID, RBAC, FHIR, HL7, T-SQL Advanced, KQL, ARM Templates, Schema Evolution, Data Quality Management

PROFESSIONAL EXPERIENCE

Senior Data Engineer

Tata Consultancy Services — EU BFSI / Enterprise

Hyderabad, India

Apr 2022 – Present

Strategic engineering leader serving as the primary onsite technical point of contact for Product Owners and Architects. Managed offshore delivery of **5+ engineers**, ensuring zero schedule slippage across sprint deliverables while building highly resilient systems.

LEADERSHIP

Served as **primary technical point of contact** for onsite Product Owners and Architects, facilitating requirement discovery, managing offshore delivery of **5+ engineers**, and ensuring **zero schedule slippage** across all sprint deliverables.

LAKEHOUSE

Designed a robust **Medallion Architecture (Bronze / Silver / Gold)** managing **5TB+ data**. Implemented idempotent ETL patterns with strict integrity constraints (MERGE INTO / ACID Delta Lake) to ensure **100% data consistency** across all pipeline retries and transient cluster failures.

FINOPS

Led compute optimization — dynamically tuning **DIUs and Parallelism** in Azure Data Factory and resolving Integration Runtime bottlenecks — delivering a **40% reduction in data processing latency** and significant cloud cost savings across enterprise pipelines.

FABRIC

Architected resilient **Microsoft Fabric semantic models using Direct Lake** mode, carefully managing F64 capacity memory guardrails (25 GB) to entirely eliminate refresh failures and ensure **high availability for C-level dashboards**, preventing DirectQuery fallback latency.

DEVOPS

Automated **CI/CD via Azure DevOps / YAML** with automated rollback strategies; enforced strict **Schema Evolution** policies to maintain 100% downstream platform stability across 5+ concurrent engineering squads.

Orchestrated enterprise data modernization pipelines, translating complex client business logic into high-efficiency Azure infrastructure.

ETL PERFORMANCE

Re-engineered legacy pipelines using **Azure Data Factory**, solving the Small File Problem and cutting query execution times by **40%** through parallelism tuning, partition optimization, and Integration Runtime right-sizing.

REPORTING

Redesigned **Power BI models with aggregation tables**, improving report load speeds by **25%** and achieving high availability across all executive and operational reporting workloads.

Led end-to-end data platform engineering and BI solution delivery, specializing in secure architecture design and transactional query tuning for financial reporting workloads.

DATA QUALITY

Improved data model accuracy by **20%** through automated data cleansing and validation frameworks, establishing consistent data quality standards across enterprise financial reporting pipelines.

SECURITY

Integrated Power BI with SaaS applications, implementing **Row-Level Security (RLS)** for sensitive financial data and designing role-based access controls governing the end-to-end lifecycle from data modeling to dashboard deployment.

PERFORMANCE

Increased data processing efficiency by **15%** through advanced T-SQL query tuning, index optimization, and stored procedure refactoring across large-scale SQL Server and Azure SQL workloads.

PLATFORM

Collaborated with cross-functional teams to develop high-impact data products utilizing **SQL Server and Azure services** for efficient large-scale processing, supporting business intelligence initiatives across multiple domains.

KEY PROJECTS & ARCHITECTURAL HIGHLIGHTS

5TB Medallion Lakehouse — EU Enterprise (TCS)

Fabric · Delta Lake · ADLS Gen2 · ADF · Azure Monitor

Designed Bronze-Silver-Gold Medallion architecture managing 5TB+ data for EU BFSI clients at TCS. Implemented idempotent MERGE INTO patterns ensuring 100% data consistency across all pipeline retries. Delivered 99.9% platform reliability with full GDPR compliance.

Microsoft Fabric Semantic Model — C-Level Dashboards

Microsoft Fabric · Direct Lake · OneLake · Power BI · DP-600

Architected resilient Fabric semantic models using Direct Lake mode at TCS. Managed F64 capacity guardrails (25 GB memory limit) to eliminate all refresh failures, accelerating C-level dashboard render speeds and achieving high availability for executive reporting.

ADF Pipeline Re-engineering — Small File Elimination (LTI)

Azure Data Factory · Synapse · Power BI · Aggregation Tables

Re-engineered legacy ETL pipelines at LTIMindtree to solve the Small File Problem, cutting query execution times by 40%. Simultaneously redesigned Power BI aggregation models, improving report load speeds by 25% across all high-availability business reports.

FinOps & Compute Optimization — DIU Tuning (TCS)

ADF · DIU Optimization · Integration Runtime · CI/CD YAML

Led Cloud FinOps initiative at TCS optimizing DIU allocation and parallelism across enterprise ADF pipelines. Resolved Integration Runtime bottlenecks and automated CI/CD via Azure DevOps YAML, delivering a 40% reduction in data processing latency with significant measurable cost savings.

EDUCATION

B.Tech – Electrical & Electronics Engineering (EEE) — 70%

Raghu Engineering College, Visakhapatnam — Aug 2013 – May 2017

Intermediate – MPC — 90%

Narayana Junior College, Vizianagaram — May 2011 – May 2013

Class 10th — 90%

New Central School, Vizianagaram — Mar 2010 – Mar 2011